📄 **Business Insights Report – Task 04**

**Exploratory Data Analysis (EDA) on Marketing Campaign Dataset**

**Data Analyst Intern – Skill Craft Technology**

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## 1️⃣ Objective

The objective of this project is to perform **Exploratory Data Analysis (EDA)** on a marketing campaign dataset to understand customer behaviour, purchasing patterns, and campaign effectiveness. The analysis aims to generate actionable business insights that support data-driven marketing decisions.

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## 2️⃣ Dataset Overview

The dataset contains customer demographic information, purchasing behaviour, and marketing campaign responses. Key attributes include:

- Customer Age and Income
- Education and Marital Status
- Product Spending Categories
- Purchase Channels (Web, Store, Catalog)
- Campaign Acceptance and Response

The dataset was cleaned by handling missing values, correcting data types, and creating additional analytical features such as **Age**, **Total Spending**, and **Total Children**.

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## 3️⃣ Exploratory Data Analysis

EDA was performed using Python to understand customer distribution, spending behaviour, and campaign performance through statistical summaries and visualizations.

Key analysis areas included:

- Customer demographic analysis
- Income and spending relationship
- Channel-wise purchase behaviour
- Campaign response evaluation
- Customer segmentation patterns

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## 4️⃣ Key Insights

- The majority of customers belong to the middle-age group, representing the primary target audience.

- Customers with higher income levels demonstrate significantly higher spending behaviour.
- Store purchases are higher compared to web and catalog purchases, indicating preference for physical shopping channels.
- Marketing campaign response rate is relatively low, suggesting improvement opportunities in targeting strategies.
- Customer characteristics such as education level and marital status influence purchasing patterns.

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## 5️⃣ Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Focus marketing campaigns on high-income customer segments to maximize revenue.
- Increase investment in high-performing purchase channels.
- Improve digital marketing strategies to enhance campaign response rates.
- Implement personalized marketing campaigns based on customer demographics and behaviour.
- Use customer segmentation to design targeted promotional offers.

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## 6️⃣ Conclusion

The Exploratory Data Analysis successfully identified important customer behaviour trends and marketing performance patterns. The insights derived from this analysis can help organizations optimize marketing strategies, improve customer engagement, and make informed business decisions.

This task demonstrates practical skills in **data cleaning, visualization, analytical thinking, and business insight generation**, reflecting real-world data analyst responsibilities.